O-P-559
26 MAY 1952

FEDERAL SPECIFICATION

POTASSIUM DICHROMATE: (POTASSIUM BICHROMATE) TECHNICAL GRADE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 **Type and grade.**—Potassium dichromate, as specified herein, shall be furnished in one type and grade.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

2.1 **Specifications.**—The following Federal Specifications, of the issues in effect on the date of invitation for bids, form a part of this specification:

P-S-536—Soap and Soap-Products (Including Synthetic Detergents); Methods of Sampling and Testing.

RR-S-366—Sieves; Standard, Testing.

2.2 **Other publication.**—The following publication of the Pharmacopoeial Convention, of the issue in effect on the date of invitation for bids, forms a part of this specification:

Pharmacopoeia of the United States.

(The Pharmacopoeia of the United States is published by the Mack Publishing Company, Easton, Pa.)

2.3 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Potassium dichromate shall be in crystalline form, shall not be caked in the container, and shall comply with the following requirements.

3.1.1 *Potassium dichromate ($K_2Cr_2O_7$) content.*—Shall be not less than 99.0 percent, by weight, of the sample.

3.1.2 *Volatile Matter (at 115° C. to 125° C.).*—Shall be not more than 0.2 percent, by weight.

3.1.3 *Water-insoluble matter.* — Not more than 0.1 percent, by weight shall be insoluble in water.

3.1.4 *Particle size.* — All of the material shall pass a No. 10 sieve, not more than 25 percent shall pass a No. 100 sieve.

3.1.5 *Sulfates.*—The sulfate (SO_4) content shall not exceed 0.1 percent.

3.1.6 *Chlorides.*—The chloride (Cl) content shall not exceed 0.1 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Unless otherwise specified, at least two samples from each lot of 500 units shall be taken from each delivery and forwarded to the laboratory for test. The sample shall be not less than 1 pound or more than 5 pounds.

4.2 **Tests.**

4.2.1 *Potassium dichromate* ($K_2Cr_2O_7$).—Weigh accurately approximately 4 grams of the sample, dissolve in distilled water, transfer the solution quantitatively to a 1-l. volumetric flask, and dilute to volume. Transfer a 50-ml. aliquot to a 250-ml. glass-stoppered Erlenmeyer flask. Add 4 to 5 grams reagent grade potassium iodide crystals, shake to dissolve, and add 10 ml. diluted sulfuric acid (1:3). Allow to stand for 10 minutes. Titrate the liberated iodine with approximately 0.1 normal sodium thiosulfate solution. When the iodine color is nearly discharged, add 5 ml. starch indicator test solution and continue the titration until the blue color just disappears (1 ml. 0.1 normal sodium thiosulfate solution is equivalent to 0.00490 gm. $K_2Cr_2O_7$). Correct for any thiosulfate consumed by a blank. Calculate the percent, by weight, potassium dichromate in the sample.

4.2.1.1 *Standard sodium thiosulfate solution.* — Dissolve 25 grams reagent grade sodium thiosulfate crystals ($Na_2S_2O_3 \cdot 5H_2O$) and 0.2 gm. sodium carbonate in the liter of recently boiled and cooled distilled water. Following the procedure given in 4.2.1, standardize the solution against recrystallized reagent grade potassium dichromate which has been dried to constant weight at about 120° C. The solution should be restandardized frequently.

4.2.1.2 *Starch indicator test solution.*—Starch indicator test solution shall be in accordance with the latest edition of the United States Pharmacopoeia.

4.2.2 *Volatile matter at 115° C. to 125° C.*).—Quickly crush a quantity of the sample to pass a No. 20 sieve. Weigh accurately about 5 gm. of the crushed sample into a shallow dish 6 to 8 cm. in diameter and 1 to 3 cm. in depth. Dry in an oven at 120° C. – 5° C. to constant weight. Calculate the percent loss in weight of the sample as volatile matter.

4.2.3 *Water-insoluble matter.* — Weigh accurately about 10 gm. of the sample. Dissolve in approximately 100 ml. warm distilled water and heat the solution on a steam bath for one hour. Filter through a tared filter crucible and wash the crucible and residue with warm distilled water until the washings are colorless (about 100 ml. in small portions). Dry the crucible and residue in an oven at 105° to 110° C. to constant weight. Calculate the gain in weight as percent sieve tests, of Federal Specification P–S–536.

4.2.4 *Particle size.*—The sieve tests shall be made in accordance with Method 210.1, sieve tests, or Federal Specification P–S–536.

4.2.5 *Sulfates.*—Dissolve approximately 10 gm. of sample, accurately weighed, in 600 ml. in a 1 l. beaker. Add 75 ml. ethyl alcohol and 60 ml. concentrate HCl. When the reaction has ceased, bring to a boil and concentrate to a thick syrup. Dilute with water to about 400 ml. and heat to boiling, while boiling slowly add 50 ml. hot 10% Barium Chloride solution, continue boiling for 5 minutes, then let stand overnight at 40-50° C. Filter the precipitate on a tared Gooch crucible, wash with 300 ml. boiling water, dry 30 minutes at 150° C., ignite over a Meker burner for 3 minutes, cool in a desiccator and weigh.

$$\text{gm. Barium Sulfate} \times 0.4115 \text{ gm. Sulfate } (SO_4)$$

4.2.6 *Chlorides.* — Dissolve approximately 10 gm. of sample, accurately weighed, in 50 ml. water in a 250 ml. beaker. Neutralize with a saturated C.P. solution of Sodium

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:14Z
Check the source to verify that this is the current version before use.

Carbonate until the color changes to yellow. Titrate with 0.1N standard solution of Silver Nitrate V.S. The end point is reached when the color first changes from clear yellow to muddy brown.

1 ml. 0.1N Silver Nitrate V.S. =
0.00355 gm. chloride. (Cl)

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** — Unless otherwise specified, commercial packaging is acceptable under this specification.

5.2 **Packing.**—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.**

5.3.1 *Issue packages.* — Unless otherwise specified, issue packages shall be marked with the name of the material, the type and quantity contained therein, the name or trade-mark of the contractor, and a warning label (5.3.1.1).

5.3.1.1 *Warning label.*—The following data shall be attached to each individual container:

POTASSIUM DICHROMATE
WARNING! **Harmful Dust.**
May Cause Rash or External Ulcers.
Keep container closed.
Avoid contact with skin and eyes.
Avoid breathing dust or solution spray.
In case of contact, immediately flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention.
Wash clothing before re-use.
Use fresh clothing daily. Take hot shower after work using plenty of soap.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the type and quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Purchasers should exercise any desired options offered herein. (See 4.1, 5.1, 5.2, 5.3.1, 5.3.2, and par. 7).

6.2 Potassium dichromate, as covered by this specification, is a technical grade intended for general use.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards, of the issues in effect on date of invitation for bids, and special requirements,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

form a part of this specification for purchases made under this specification by the respective departments:

7.2 **Army, Navy, and Air Force.**

7.2.1 *Applicable specifications and standards.*

Military Specifications:

JAN–D–111—Drums, Fiberboard (Overseas Type).

JAN–P–134—Packaging and Packing for Overseas Shipment—Kegs, Slack (for Kegs Whose Weight of Contents Does Not Exceed 250 Pounds).

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking of Shipments.

7.2.2 *Inspection procedures (Navy only).*—The general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.3 *Sampling procedures.* — Sampling shall be in accordance with Military Standard MIL–STD–105. The acceptable quality level (AQL) shall be 2.5 percent defective. The inspection level shall be level I.

7.2.4 *Packing.*

7.2.4.1 *For domestic shipment.*—Potassium dichromate, technical grade, shall be packed in lots of 100 pounds. Containers shall be either type I fiberboard drums with all fiber or plywood heads, or slack kegs conforming to the requirements of Military Specifications JAN–D–111 and JAN–P–134, respectively. Slack kegs shall be lined with a material which shall not affect nor be affected by the product to be contained nor will permit sifting of the contents from the keg.

7.2.4.2 *For overseas shipment.* — Packing shall be as specified in 7.2.4.1 but, in addition, strapping shall be applied to kegs as required by Military Specification JAN–P–134.

7.2.5 *Marking.*—Shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

7.2.6 *Ordering data.* — Procurement documents should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment (see 7.2.4 and 7.2.5).

7.2.7 This specification covers the following item appearing in the Armed Services Catalog of Medical Materiel:

1–363–000—Potassium Dichromate, ¼ lb.

NAVY INTEREST: A Md Or Sh S Y

[1] Applicable to Navy purchases only.

(Activities outside the Federal Government may obtain copies of Federal specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:14Z
Check the source to verify that this is the current version before use.